

Multiple Choice

1. **Answer:** C

Options: A) `cmp(list)`; B) `len(list)`; C) `max(list)`; D) `min(list)`

Note: The function '`max(list)`' in Python 3 returns the item from the list that has the highest value, making it the correct choice for identifying the maximum element in a list.

2. **Answer:** C

Options: A) `[19,21]`; B) `[11,15]`; C) `[11,15,19]`; D) `[13,17,21]`

Note: The correct answer is `[11,15,19]` because the slice notation `b[::2]` extracts every second element from the list `b`, starting from the first element at index 0.

3. **Answer:** A

Options: A) `**`; B) `//`; C) `is`; D) `not in`

Note: The operator '`**`' in Python 3 is specifically designed to perform exponentiation, allowing users to raise a number to the power of another number.

4. **Answer:** C

Options: A) `17_{16}`; B) `E4_{16}`; C) `E7_{16}`; D) `F4_{16}`

Note: The correct answer is `E7_{16}` because when converting the decimal number 231 to hexadecimal, it equals $14 \cdot 16^1 + 7 \cdot 16^0$, which corresponds to the hexadecimal digits E (for 14) and 7.

5. **Answer:** C

Options: A) $O(N^{(1/2)})$; B) $O(N^{(1/4)})$; C) $O(N^{(1/N)})$; D) $O(N)$

Note: The function $O(N^{(1/N)})$ grows the slowest among common complexity classes because as N increases, $N^{(1/N)}$ approaches 1, indicating that it increases at a rate that diminishes significantly compared to polynomial or exponential functions.

True / False

6. **Answer:** True

Options: A) True; B) False

Note: The addition of the two's-complement numbers 10000001 (which represents -127) and 10101010 (which represents -86) results in a sum of 10101011 (which represents -53), but since the signs of the operands were both negative and the result is positive, it indicates an overflow occurred.

7. **Answer:** True

Options: A) True; B) False

Note: In Python, the multiplication operator (`*`) when applied to a list creates a new

list that contains the original list's elements repeated the specified number of times, so `['Hi!'] * 4` produces a list with four instances of 'Hi!'.

8. **Answer:** True

Options: A) True; B) False

Note: The statement is true because the left bitwise shift operator '`<<`' shifts the bits of the number 1 three positions to the left, resulting in the binary representation of 8 (which is 1000 in binary).

9. **Answer:** True

Options: A) True; B) False

Note: In Python, tuples are zero-indexed, meaning that the first element can be accessed using index 0, which in this case is 'abcd'.

10. **Answer:** False

Options: A) True; B) False

Note: The Caesar Cipher is not perfectly secure because it relies on a simple substitution method that can be easily broken through frequency analysis and brute force attacks, as there are only a limited number of possible keys (25).

Long Form Response

11. **Answer:** `def most_common_elem(s,a): | return most_common_elem`

Note: The answer correctly identifies the need to define a function that processes a given text string to determine the frequency of each element, ultimately returning the most common elements along with their counts, which is a fundamental requirement of the problem.

12. **Answer:** `def remove_words(list 1, charlist): | return new_list`

Note: correct because it outlines a function that takes a list of strings and a character or string to filter out, implying that it will create a new list without the specified words, though the implementation details are incomplete.

End of Answer Key